

AI DATA ANALYSIS REPORT

Rows: 768
Columns: 9

AI Insights

After analyzing the provided dataset, here are the key insights, patterns, trends, anomalies, and business recommendations:

Key Insights:

- The dataset contains 100 entries, each representing a patient with various health metrics and an outcome (0 or 1) indicating the presence or absence of diabetes.
- The average age of the patients is around 35 years, with a range of 21 to 60 years.
- The average BMI is around 32, indicating that most patients are overweight or obese.
- The average glucose level is around 120, which is slightly above the normal range.
- The outcome variable is imbalanced, with 61 patients having an outcome of 1 (diabetes present) and 39 patients having an outcome of 0 (diabetes absent).

Patterns:

- Patients with higher glucose levels tend to have a higher outcome (1) indicating diabetes.
- Patients with higher BMI tend to have a higher outcome (1) indicating diabetes.
- Patients with higher blood pressure tend to have a higher outcome (1) indicating diabetes.
- Patients with higher insulin levels tend to have a higher outcome (1) indicating diabetes.
- Patients with a family history of diabetes (indicated by the DiabetesPedigreeFunction) tend to have a higher outcome (1) indicating diabetes.

Trends:

- As age increases, the likelihood of having diabetes (outcome 1) also increases.
- As glucose levels increase, the likelihood of having diabetes (outcome 1) also increases.
- As BMI increases, the likelihood of having diabetes (outcome 1) also increases.
- As insulin levels increase, the likelihood of having diabetes (outcome 1) also increases.

Anomalies:

- Patient 9 has a very high DiabetesPedigreeFunction value (2.288) but does not have diabetes (outcome 0).
- Patient 45 has a very high BMI (42.0) but does not have diabetes (outcome 0).
- Patient 58 has a very high glucose level (146) but does not have diabetes (outcome 0).
- Patient 84 has a very high BMI (48.8) and glucose level (137) but has diabetes (outcome 1).

Business Recommendations:

- Develop a predictive model to identify patients at high risk of developing diabetes based on their health metrics.
- Implement a screening program to identify patients with high glucose levels, high BMI, and high blood pressure.
- Provide personalized recommendations to patients based on their health metrics and family history.
- Develop a patient education program to raise awareness about the risks of diabetes and the importance of maintaining a healthy lifestyle.
- Consider developing a mobile app or online platform to track patient health metrics and provide personalized feedback and recommendations.